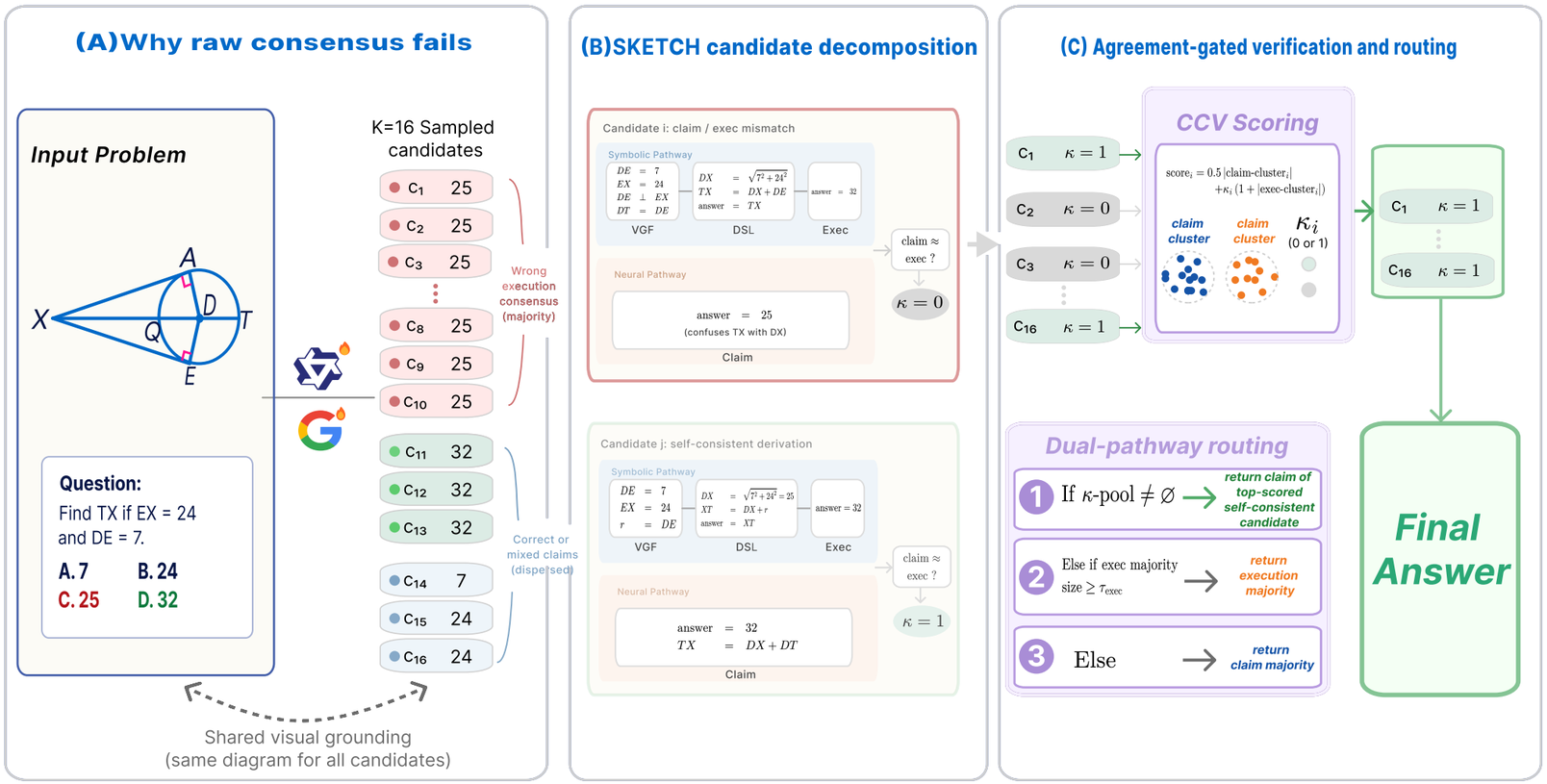

(A)Why raw consensus fails
Input Problem
A
X
Q
D
T
E
Question:
Find TX if EX = 24
and DE = 7.
A. 7
B. 24
C. 25
D. 32
K=16 Sampled candidates
C1 25
C2 25
C3 25
C8 25
C9 25
C10 25
C11 32
C12 32
C13 32
C14 7
C15 24
C16 24
Wrong execution consensus (majority)
Correct or mixed claims (dispersed)
Shared visual grounding
(same diagram for all candidates)
(B)SKETCH candidate decomposition
Candidate i: claim / exec mismatch
Symbolic Pathway
DE = 7
EX = 24
DE ⊥ EX
DT = DE
DX = √(7² + 24²)
TX = DX + DE
answer = TX
answer = 32
VGF
DSL
Exec
Neural Pathway
answer = 25
(confuses TX with DX)
Claim
claim ≈ exec ?
κ = 0
Candidate j: self-consistent derivation
Symbolic Pathway
DE = 7
EX = 24
r = DE
DX = √(7² + 24²) = 25
XT = DX + r
answer = XT
answer = 32
VGF
DSL
Exec
Neural Pathway
answer = 32
TX = DX + DT
Claim
claim ≈ exec ?
κ = 1
(C) Agreement-gated verification and routing
C1 κ = 1
C2 κ = 0
C3 κ = 0
C16 κ = 1
CCV Scoring
score_i = 0.5 |claim-cluster_i| + κ_i (1 + |exec-cluster_i|)
claim cluster
claim cluster
κ_i
(0 or 1)
C1 κ = 1
C16 κ = 1
Dual-pathway routing
1 If κ-pool ≠ ∅ → return claim of top-scored self-consistent candidate
2 Else if exec majority size ≥ τ_exec → return execution majority
3 Else → return claim majority
Final Answer